

Education

- 2024 - 2026 **M.Sc. in Physics**, **CREATE Scholar**, *University of Victoria*, Victoria, Canada.
◇ **Thesis Advisor:** Prof. Thomas Baker.
◇ **Thesis Title:** Scaling Analysis and Ensemble Measures in Quantum Systems.
- 2019 - 2023 **B.Sc. in Honours Physics**, *McGill University*, Montréal, Canada.
◇ **Thesis Advisor:** Prof. Keshav Dasgupta.
◇ **Thesis Title:** Conflicts with de Sitter vacua in supersymmetric field theories.

Research Interests

Quantum information and communication; quantum algorithms; quantum sensing; numerical quantum many-body physics; high-performance computing; quantum chemistry.

Technical Expertise

- Programming Python, Julia, Bash, TypeScript, Solidity.
- Scientific / NumPy, SciPy, Qiskit, PennyLane; exact diagonalization, tensor networks, numerical linear algebra,
Quantum Monte Carlo methods.
- Software / Linux, Git, pytest, CI/CD, HPC cluster workflows, parallel numerical computing, scientific visualiza-
HPC tion.

Publications

- 2025 J. Morais, *Conflicts with de Sitter Vacua in Superstring Theory*, *McGill Science Undergraduate Research Journal* **20(1)** (2025), pp. 59–64.

Research Experience

- 2024 - 2026 **Graduate Researcher**, *University of Victoria*, Victoria, Canada
Computational study of thermalization and the eigenstate thermalization hypothesis across integrable and non-integrable quantum many-body models using exact diagonalization and tensor-network methods. Developed and validated numerical pipelines for level-spacing statistics, ensemble measures, relaxation dynamics, and finite-size scaling across rapidly growing Hilbert spaces, including large-scale execution on Linux HPC systems. Contributed to development of **DMRJtensor.jl**, a general-purpose tensor-network library in Julia.
Supervisor: Prof. Thomas Baker
- 2023 - 2026 **Quantum Software Lead**, *BTQ*, Vancouver, Canada
Developed research software and algorithms spanning quantum error-correction tooling, cryptography, distributed consensus, and quantum random number generation. Led four BTQ staff and two M.Sc. researchers across concurrent quantum-software projects, including **QLDPC**, an interactive Python toolkit for quantum LDPC and surface-code construction, Tanner-graph visualization, Qiskit circuit generation, and belief-propagation decoding, and **QRiNG**, a Python prototype for quantum randomness and consensus validation with smart-contract components. Built and validated software with emphasis on testing, reproducibility, scaling, and maintainability.  
Collaborator: Prof. Gavin Brennen
- 2023 - 2024 **Researcher**, *Fudan University*
Studied generalization in topological quantum neural networks using topological quantum field theory and spin-network encodings. Developed a Python research toolkit combining tensor-network simulation, noise-resilient classification, automated testing, and interactive visualization. 
Supervisors: Prof. Antonino Marcianò, Prof. Emanuele Zappala

- Summer 2023 **Undergraduate Researcher**, *NSERC, University of Alberta*, Edmonton, Canada
 Studied the extended entanglement structure of entangled qubit systems with non-perturbative topological wormhole corrections. Used these corrections to characterize tunneling structure and explore quantum-algorithmic applications in lattice-confined qubit systems.
 Supervisor: Prof. Igor Boettcher
- 2022 - 2023 **Honours Bachelor Thesis**, *McGill University*, Montréal, Canada
 Investigated the absence of vacua with de Sitter isometries in type II string theory and M-theory. Studied generalized coherent states over supersymmetric Minkowski space with these isometries as a route to non-singular compactification to de Sitter spacetime, a candidate model for our Lorentzian universe.
 Supervisor: Prof. Keshav Dasgupta
- Summer 2022 **Undergraduate Researcher**, *NSERC, McGill University*, Montréal, Canada
 Characterized U(1) topological-defect (cosmic-string) signatures in a class of renormalizable quantum field theories. Developed signal-extraction methods for simulated primordial 21-cm Λ CDM backgrounds, including wavelet and matched-filter analysis. 🐱
 Supervisor: Prof. Robert Brandenberger
- Spring 2022 **Undergraduate Researcher**, *McGill University*, Montréal, Canada
 Studied the interaction of light and the dynamical Casimir effect in photon recycling using scalar quantum field theory. Computed quantum corrections to the radiation force for relativistically moving mirror boundary conditions.
 Supervisors: Prof. Simon Caron-Huot, Prof. Andrew Higgins
- Fall 2021 **Undergraduate Researcher**, *McGill University*, Montréal, Canada
 Numerically analyzed scintillation densities for fast radio bursts and their corresponding black-hole–white-hole tunneling events in quantum cosmology. The fast radio burst signals were embedded in nonlinear cosmological noise generated by primordial perturbations. 🐱
 Supervisor: Prof. Victoria Kaspi
- Summer 2021 **Undergraduate Researcher**, *McGill University Health Center*, Montréal, Canada
 Constructed and trained neural-network models for radiation dose-volume tensor estimation under helical delivery geometry.
 Supervisor: Prof. Marija Popovic
- Summer 2020 **Undergraduate Researcher**, *SURA, McGill University*, Montréal, Canada
 Analyzed γ -ray and Cherenkov emission in superluminous supernovae and tidal disruption events with the NASA Fermi-LAT, including binned/unbinned likelihood analyses, upper-limit analyses, and extended-source analyses.
 Supervisor: Prof. Kenneth Ragan
- 2018 - 2019 **Undergraduate Researcher**, *Vanier College*, Montréal, Canada
 Numerically solved quantum Hamilton–Jacobi equations and generated de Broglie–Bohm trajectories using recurrent neural networks and the Crank–Nicolson method. 🐱
 Supervisor: Prof. Ivan Ivanov
- Summer 2018 **Undergraduate Researcher**, *Concordia University*, Montréal, Canada
 Studied topological confinement in a nanobeam microcavity. Characterized resonant electromagnetic modes in nanoscale photonic-crystal ring resonators using MIT Electromagnetic Equation Propagation (MEEP).
 Supervisor: Prof. Pablo Bianucci

Awards and Distinctions

- Fall 2024 **NSERC CREATE - Quantum Computing Program Scholarship** (\$13,000 CAD), *University of Victoria, Department of Physics*.
- Fall 2024 **BCGS - British Columbia Graduate Scholarship** (\$17,500 CAD), *University of Victoria, Department of Physics*.
- 2024 - 2025 **UVic FGS - University of Victoria Fellowship - Master's** (\$4,000 CAD total), *University of Victoria, Department of Physics*.

- May 2023 **NSERC USRA - Undergraduate Student Research Award + FRQNT Scholarship Supplement** (\$7,000 NSERC + \$1,500 FRQNT), *University of Alberta, Department of Physics.*
- May 2022 **NSERC USRA - Undergraduate Student Research Award + FRQNT Scholarship Supplement** (\$7,000 NSERC + \$1,500 FRQNT), *McGill University, Department of Physics.*
- May 2022 **SURE - Summer Undergraduate Research in Engineering Award (Declined)** (\$6,000 CAD), *McGill University, Department of Mechanical Engineering.*
- May 2021 **BSA - Banner Student Award** (\$4,300 CAD), *McGill University Faculty of Medicine, Medical Physics Unit.*
- May 2020 **SURA - Science Undergraduate Research Award** (\$7,000 CAD), *McGill University, Department of Physics.*

Presentations

- Feb 2025 **Quantum Days**, *University of Toronto, Toronto, Canada*
Presented work on topological approaches to cryptographic protocols for distributed consensus networks.
- July 2023 **8th Interstellar Symposium**, *McGill University, Montréal, Canada*
Presented the effects of light interference and the dynamical Casimir effect in photon recycling via scalar quantum field theory.
- May 2019 **Physics & AI Workshop**, *McGill University, Montréal, Canada*
Presented numerically computed quantum Hamilton–Jacobi trajectories for de Broglie–Bohm theory using recurrent neural networks and the Crank–Nicolson method. 🐱

Selected Academic Activities

- July 2025 **Qiskit Summer School**, *Quantum Computing*, Participated in IBM’s Qiskit Summer School on advanced quantum algorithms and quantum machine learning for near-term quantum devices.
- May 2025 **Quantum Chemistry Workshop**, *Variational Quantum Eigensolver*, Developed VQE tooling for molecular ground-state-energy calculations, including optional visualization tools; 2nd-place finalist at the workshop hosted by the [Institut Quantique](#). 🐱
- 2022 - 2023 **Group Seminar**, *Superstring Theory*, Organized a graduate seminar on superstring theory based on modern HEP research papers and textbooks by Kiritsis and Polchinski.
- Spring 2022 **Group Seminar**, *Gauge Theory*, Organized a graduate seminar on non-abelian gauge theory based on Baez’s *Gauge Fields, Knots and Gravity* and Kibble’s *Classification of Topological Defects and Their Relevance to Cosmology*.
- Fall 2021 **Physics Hackathon**, *McGill University, Montréal, Canada*
Numerically reproduced the double-slit interference pattern with path integrals using the Metropolis–Hastings algorithm and Glauber dynamics for Markov chain Monte Carlo. 🐱 🐱